

OLA Ride Booking Analytics

— Project Documentation

1. Project Title

OLA Ride Booking Analytics

2. Project Objective

Explain the business problem in 3–5 sentences.

Example:

The objective of this project is to analyze ride-booking data and identify patterns in booking success, cancellations, customer behavior, vehicle performance, ratings, payment methods, and booking value. Python is used for data inspection and preparation, PostgreSQL is used for structured SQL analysis, and Power BI is used to present the results through an interactive dashboard.

3. Business Requirements

Document the questions the project is designed to answer.

- Successful booking analysis
- Average ride distance by vehicle type
- Customer cancellation analysis
- Top customers by ride count
- Driver cancellation analysis
- Driver rating analysis
- Payment-method analysis
- Customer rating analysis
- Successful booking-value analysis
- Incomplete ride analysis

4. Dataset Information

Item

Details

Item	Details
Source file	Bookings-100000-Rows.xlsx
Format	Excel
Main table	bookings
Main analysis tools	Python, PostgreSQL, Power BI

Columns

Document each important column:

Column	Description	Data Type
date	Booking date/time	datetime
time	Booking time	time/text
booking_id	Unique booking identifier	text
booking_status	Status of the booking	categorical
customer_id	Customer identifier	text
vehicle_type	Type of vehicle booked	categorical
pickup_location	Pickup location	categorical
drop_location	Drop location	categorical
v_tat	Vehicle turnaround/arrival metric	numeric
c_tat	Customer turnaround/arrival metric	numeric
canceled_rides_by_customer	Customer cancellation reason	categorical
canceled_rides_by_driver	Driver cancellation reason	categorical
incomplete_rides	Indicates incomplete ride	categorical
incomplete_rides_reason	Reason for incomplete ride	categorical
booking_value	Booking amount/value	numeric
payment_method	Payment method	categorical
ride_distance	Ride distance	numeric
driver_ratings	Driver rating	numeric
customer_rating	Customer rating	numeric
vehicle_images	Vehicle image URL	text

5. Data Preparation

Mention:

- File loading
- Column-name standardization
- Data-type checks
- Missing-value investigation
- Duplicate checks
- Invalid-value checks
- Outlier checks
- Categorical-value validation

- Date/time conversion
- Numeric-column validation

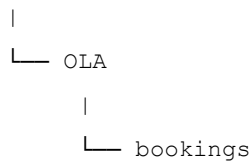
Example code:

```
df.columns = df.columns.str.lower()  
df.columns = df.columns.str.replace(" ", "_")
```

6. Database Design

Database

PostgreSQL



Document:

- Database name
- Table name
- Number of columns
- Primary key, if defined
- Important categorical columns
- Important numeric columns

7. Python Workflow

Document the notebook flow:

```
Read Excel
↓
Inspect DataFrame
↓
Check data types
↓
Check missing values
↓
Standardize column names
↓
Connect to PostgreSQL
↓
Load bookings table
```

8. SQL Analysis

Document every query/view using this format:

Analysis 1 — Successful Bookings

Business question:

How many/which bookings were successful?

SQL view:

```
successful_bookings
```

Logic:

```
Filter booking_status = 'Success'.
```

Business use:

Used to analyze completed/successful ride activity.

Repeat this format for all 10 SQL analyses.

9. Power BI Dashboard

Document:

Page 1 — Overview

- KPI cards
- Total bookings
- Successful bookings

- Cancelled bookings
- Booking value
- Average rating

Page 2 — Vehicle Analysis

- Bookings by vehicle type
- Average ride distance
- Average customer rating
- Booking value by vehicle type

Page 3 — Cancellation Analysis

- Customer cancellations
- Driver cancellations
- Cancellation reasons
- Vehicle type cancellation patterns

Page 4 — Customer Analysis

- Top customers
- Customer ratings
- Ride frequency
- Payment-method usage

10. KPI Definitions

Create a table like:

KPI	Definition
Total Bookings	Count of booking records
Successful Bookings	Count where booking status is Success
Cancelled Bookings	Count of cancelled booking records
Total Booking Value	Sum of booking value
Average Ride Distance	Average ride distance
Average Customer Rating	Average customer rating
Average Driver Rating	Average driver rating

11. Business Insights

Do not write generic statements.

For every insight, include:

Observation → Evidence → Business implication

Example:

Observation: *A particular vehicle type has a higher average customer rating.*

Evidence: *The Power BI analysis shows the difference in average rating by vehicle type.*

Business implication: *The company could investigate what operational factors contribute to the higher customer satisfaction.*

12. Recommendations

Recommendations should be connected to actual findings.

Examples:

- Investigate the most common driver cancellation reasons.
- Improve operational handling of incomplete rides.
- Study high-frequency customers for retention opportunities.
- Compare vehicle types using both demand and customer-rating metrics.
- Investigate payment-method preferences.
- Analyze successful booking value alongside ride distance.

13. Technical Architecture

```
Excel
  ↓
Python / Pandas
  ↓
PostgreSQL
  ↓
SQL Views
  ↓
Power BI
  ↓
Business Insights
```

14. Repository Structure

```
OLA-Ride-Booking-Analytics/  
|  
├─ data/  
|   └─ Bookings-100000-Rows.xlsx  
|  
├─ notebooks/  
|   └─ main.ipynb  
|  
├─ sql/  
|   └─ OLA_DATA.sql  
|  
├─ powerbi/  
|   └─ OLA_Data.pbix  
|  
├─ README.md  
├─ PROJECT_DOCUMENTATION.md  
└─ .gitignore
```

15. Security

Never upload:

- PostgreSQL passwords
- API keys
- .env files containing credentials
- Personal credentials
- Private database connection strings

Use:

```
import os  
  
password = os.getenv("POSTGRES_PASSWORD")
```

and add:

```
.env  
*.env
```

to .gitignore.

16. Limitations

Document anything that may affect interpretation:

- Dataset coverage period
- Missing values
- Data-quality limitations
- Assumptions
- Whether the dataset represents real or simulated business data
- Power BI refresh requirements
- Local PostgreSQL dependency

17. Future Scope

- Automated ETL pipeline
- Cloud database
- Scheduled Power BI refresh
- Advanced DAX measures
- Data warehouse/star schema
- More advanced SQL analytics
- Predictive analysis
- Customer segmentation
- Cancellation prediction